Data Visualization

Data Preprocessing

Preprocessing in the Visualizing Housing Market Trends project involves cleaning and preparing raw housing data before analysis. It includes removing missing values, correcting errors, converting data formats, and handling outliers. This process ensures accurate and reliable data for creating

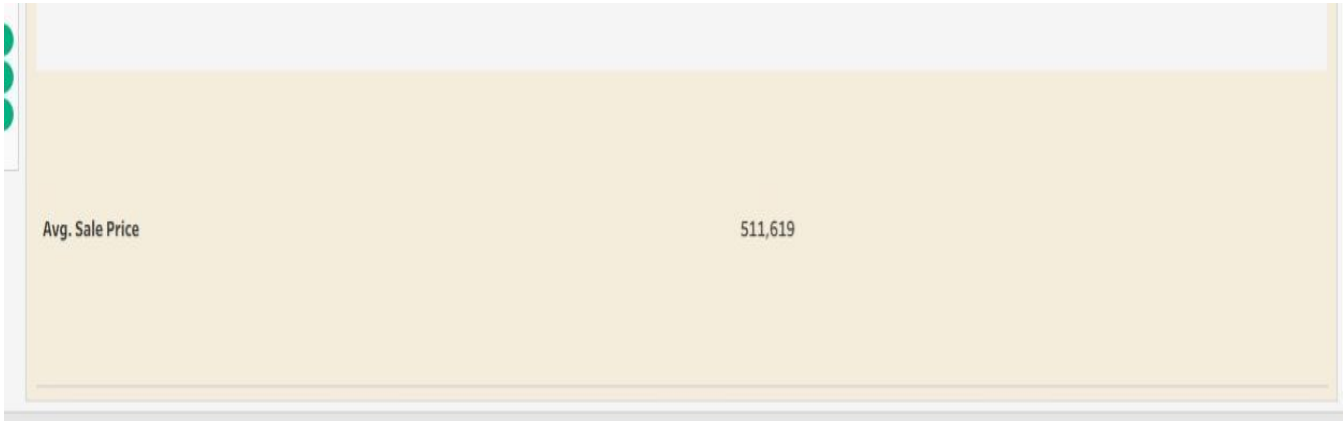
Steps:

- Data Collection – Gather housing data from reliable sources (sales, prices, locations, dates).
- Data Cleaning – Remove duplicates, missing values, and incorrect entries.
- Data Formatting – Convert data types (dates, prices, numeric fields).
- Handling Missing Values – Fill, remove, or replace null values appropriately.
- Outlier Detection – Identify and treat extreme property prices or unusual records.
- Data Integration – Merge multiple datasets (location, property type, sales data).
- Feature Engineering – Create new fields like price per sq.ft., yearly growth rate.
- Data Filtering – Remove irrelevant columns and keep necessary attributes.
- Data Aggregation – Summarize data by year, city, or property type.
- Final Validation – Verify data accuracy before building dashboards and visualizations.

Business Questions With Visualizations

- **What is the average sales price of houses?**

Avarage Sales price

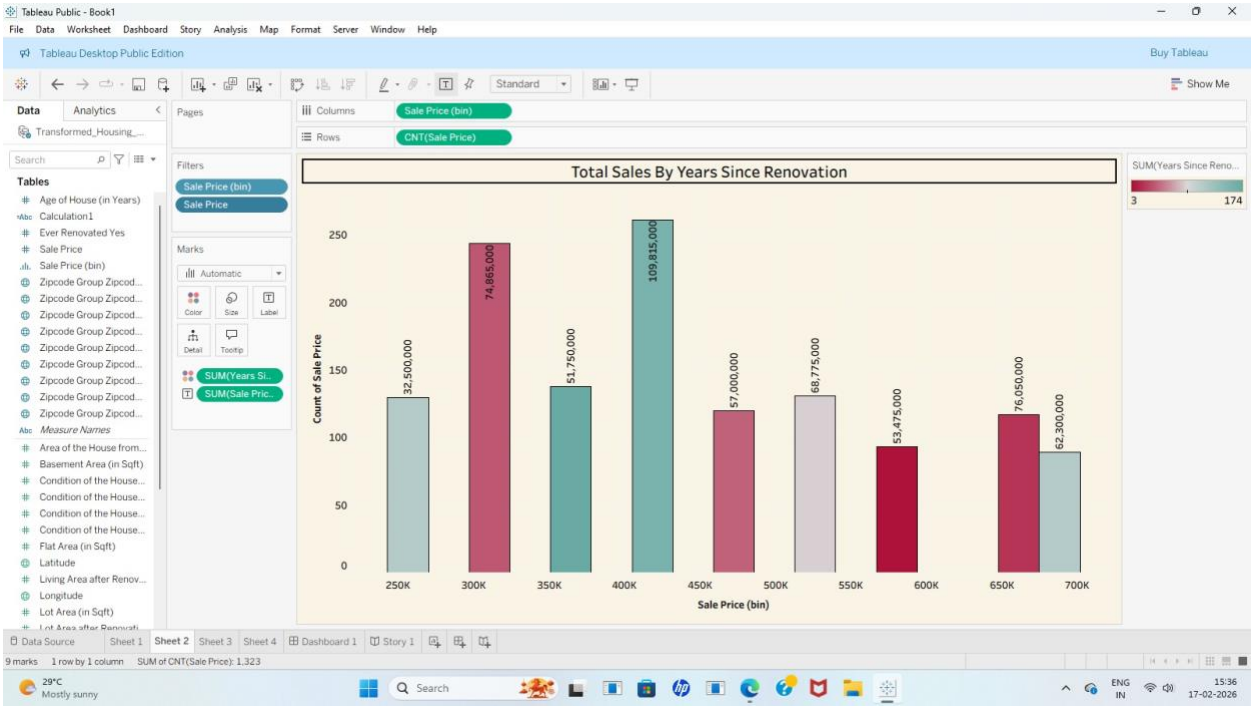


- **How many houses are included in the dataset?**

Count of houses

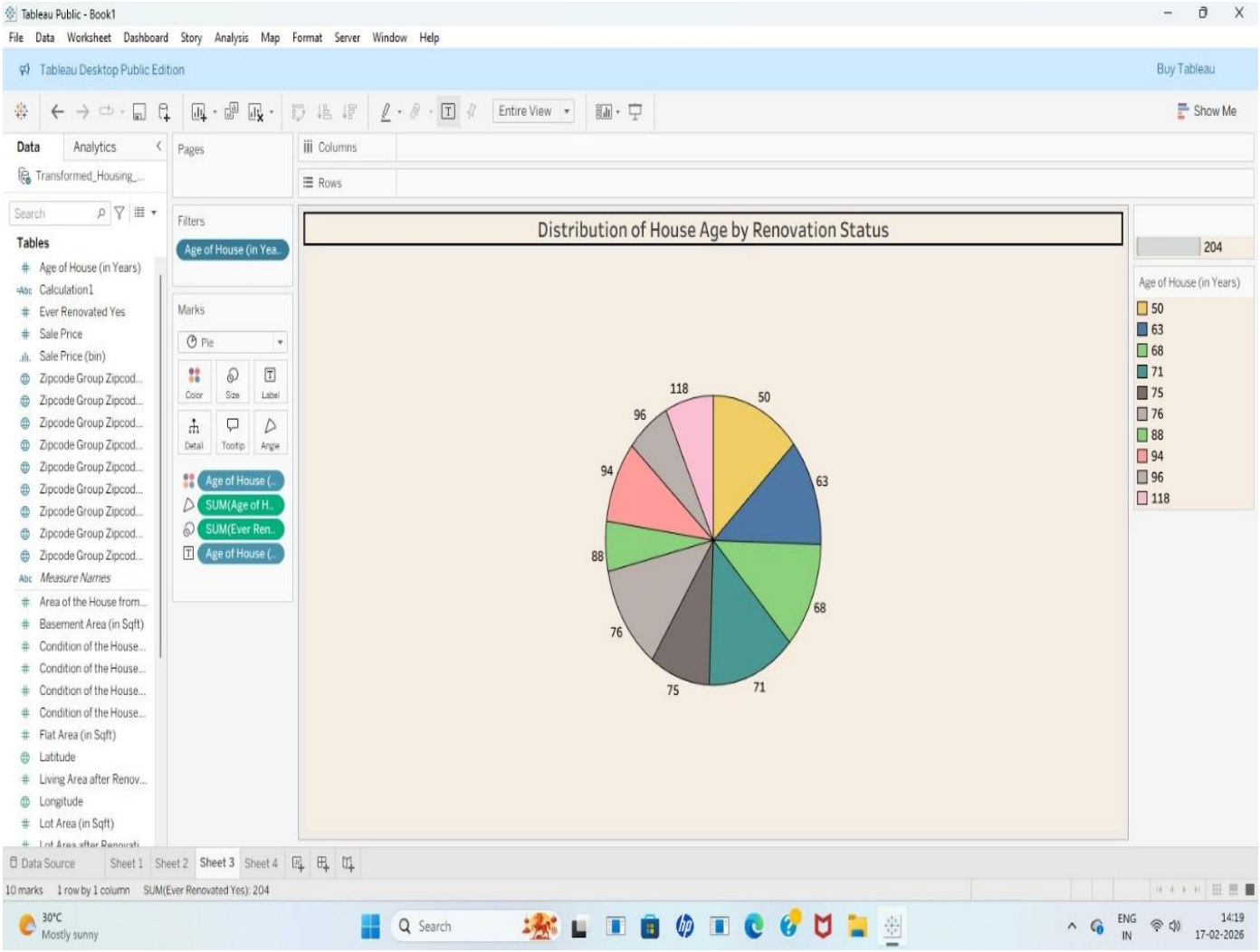
- **Do renovated houses generate more sales over time?**

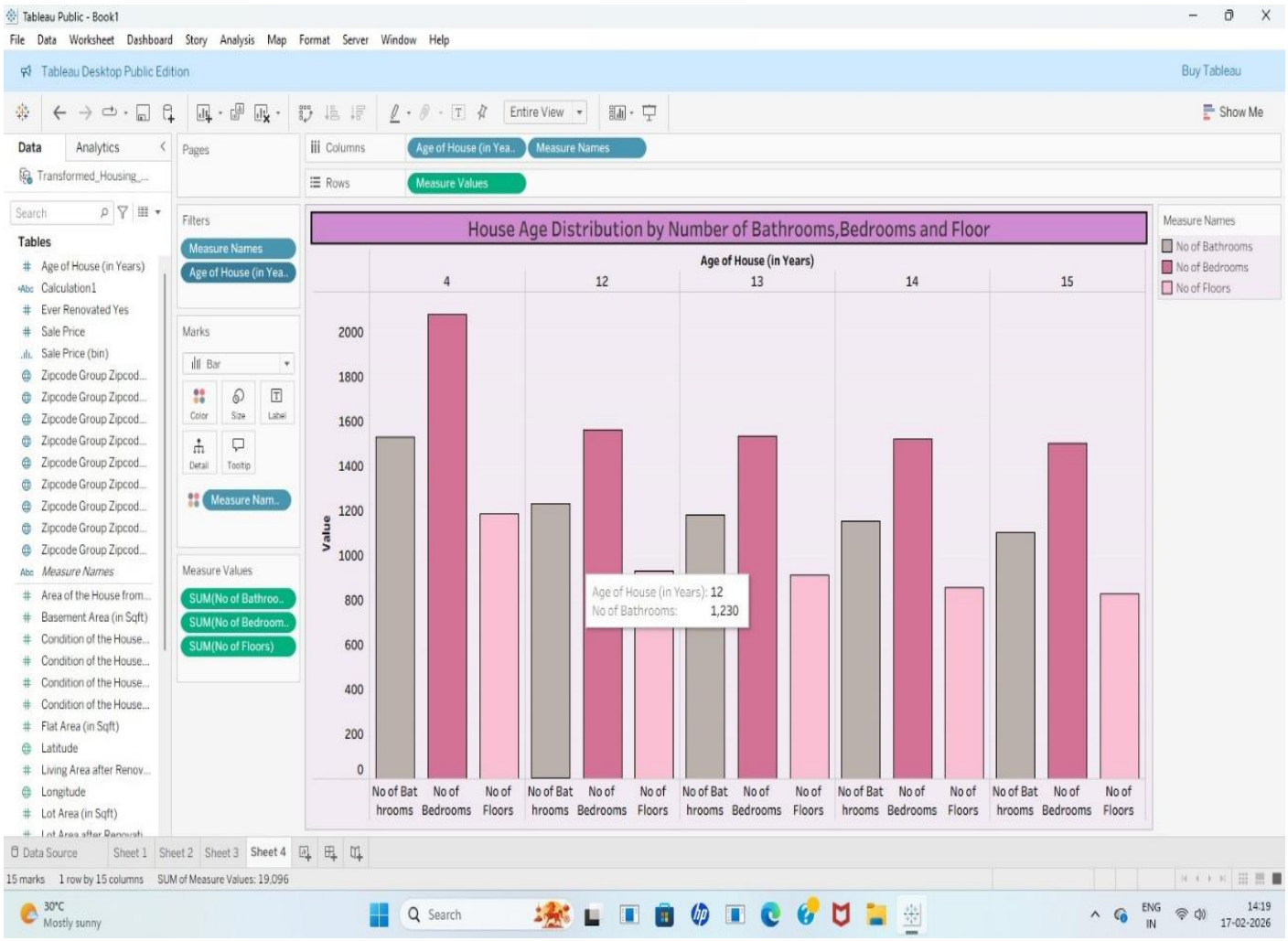
Total Sales by Year Renovation



- **Are older houses more likely to be renovated?**

Distribution of House Age & Renovation Status



- **What is the age distribution of houses?**

House Age Distribution